

Infinite Series

An infinite series is the sum of infinite sequence of numbers

$$a_1 + a_2 + a_3 + \dots + a_n + \dots$$

Since there are infinitely many terms to add in an infinite series, we cannot just keep adding to see what comes out. Instead we look at the results of summing just the first n terms of the sequence. The sum of the first n terms

$$a_1 + a_2 + a_3 + \dots + a_n$$

is an ordinary finite sum and can be calculated by normal addition. It is called the n th partial sum and it is denoted by S_n .

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$

Definition:- Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \dots + a_n + \dots$$

is an infinite series. The number a_n is the n th term of the sequence. The sequence $\{S_n\}$ defined by

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$\vdots \quad \vdots$$

$$s_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$$

is the sequence of partial sums of the series, the number s_n being the n th partial sum. If the sequence of partial sums converges to a limit L , we say that the series converges and that its sum is L . In this case, we also write

$$a_1 + a_2 + a_3 + \dots + a_n + \dots = \sum_{n=1}^{\infty} a_n = L$$

If the sequence of partial sums of the series does not converge, we say that the series diverges.

For example, an expression like

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

<u>partial sum</u>	<u>value</u>	
first! $s_1 = 1$	1	$(2 - 1)$
second! $s_2 = 1 + \frac{1}{2}$	$\frac{3}{2}$	$(2 - \frac{1}{2})$
third! $s_3 = 1 + \frac{1}{2} + \frac{1}{4}$	$\frac{7}{4}$	$(2 - \frac{1}{4})$
$\vdots$		
n th! $s_n = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}}$	$\frac{2^n - 1}{2^{n-1}}$	$(2 - \frac{1}{2^{n-1}})$

$$S_n = 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n}$$

$$= 2 - \frac{1}{2^n - 1}$$

This sequence of partial sums converges to 2 because $\lim_{n \rightarrow \infty} \frac{1}{2^n - 1} = 0$.

We say "the sum of the infinite series $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n} + \dots$ is 2."

Is the sum of any finite number of terms in this series equal to 2? No

Can we actually add infinite number of terms one by one? No

But we can still define their sum by defining it to be the limit of the sequence of partial sums as $n \rightarrow \infty$, in this case 2.

① Find the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} + \dots$$

Sol Here $a_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$

$$a_1 = \frac{1}{1} - \frac{1}{2}$$

$$a_2 = \frac{1}{2} - \frac{1}{3}$$

$$\vdots$$

$$a_n = \frac{1}{n} - \frac{1}{n+1}$$

$$S_1 = a_1 = \frac{1}{1} - \frac{1}{2}$$

$$S_2 = a_1 + a_2 = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3}$$

$$\vdots$$

$$S_n = a_1 + a_2 + \dots + a_n = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1}$$

$$= 1 - \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1 \text{ (finite)}$$

Since the n th partial sum converges and hence the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges and it converges to 1.

We say, its sum $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$.

* We can represent each term in an infinite series by the area of a rectangle. If all the terms a_n in the series are positive, then the series converges if the total area is finite, and diverges otherwise.

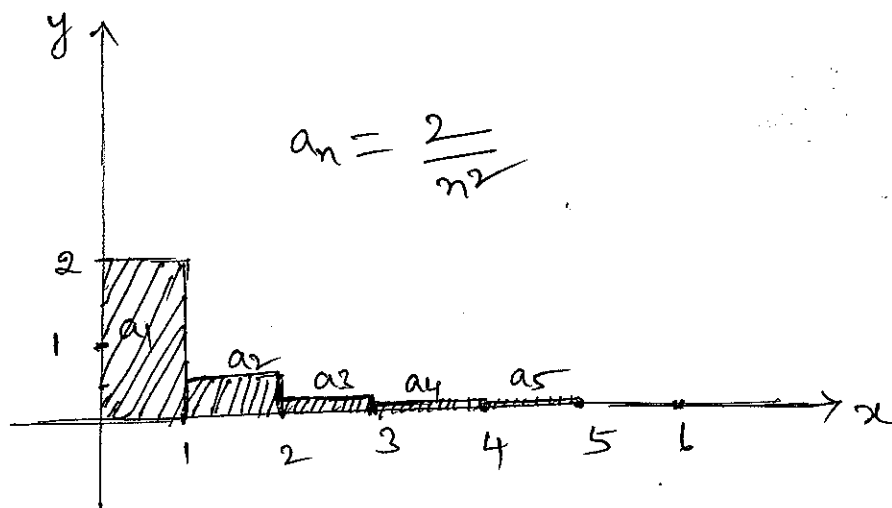


Figure: (a)

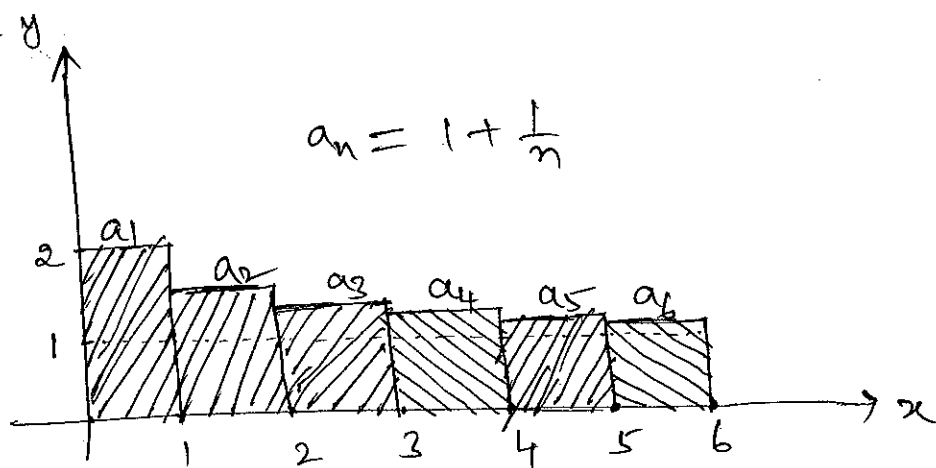


Figure: (b)

The sum of a series with positive terms can be interpreted as a total area of an infinite collection of rectangles. The series converges when the total area of the rectangles is finite (figure (a)) and diverges when the total area is unbounded (figure (b)).

Note that the total area can be infinite even if the area of the rectangles is decreasing.

* The infinite series $a_1 + a_2 + a_3 + \dots + a_n + \dots$ is usually denoted by $\sum_{n=1}^{\infty} a_n$ or more briefly, by $\sum a_n$.

Example ① Show that the series $1^2 + 2^2 + 3^2 + \dots + n^2 + \dots$ diverges to $+\infty$.

Sol: The n th partial sum $S_n = a_1 + a_2 + \dots + a_n$

$$= 1^2 + 2^2 + 3^2 + \dots + n^2$$
$$= \frac{n(n+1)(2n+1)}{6}$$
$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6} = \infty$$

$\Rightarrow \{S_n\}$ diverges to $+\infty$

$\Rightarrow$ The given series diverges to $+\infty$.

Example ② Show that the series $-1 - 2 - 3 - \dots - n - \dots$ diverges to $-\infty$.

Sol: $S_n = -1 - 2 - 3 - \dots - n$

$$= -(1 + 2 + 3 + \dots + n)$$
$$= -\frac{n(n+1)}{2}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} -\frac{n(n+1)}{2} = -\infty$$

$\Rightarrow \{S_n\}$ diverges to $-\infty$

$\Rightarrow$ The given series diverges to $-\infty$.

Example ② Test the convergence of $\sum_{n=0}^{\infty} (-1)^n$.

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Sol:-

$$S_n = 1 - 1 + 1 - 1 + 1 - 1 + \dots \text{ upto } n \text{ terms} \\ = 1 \text{ or } 0 \text{ according as } n \text{ is odd or even.}$$

The subsequence $\{S_{2n}\}$ converges to 0 while the subsequence $\{S_{2n+1}\}$ converges to 1.

$\Rightarrow \{S_n\}$ is not convergent

$\Rightarrow \{S_n\}$ is divergent

$\therefore$ the series $\sum_{n=0}^{\infty} (-1)^n$ diverges.

In particular, the series $\sum_{n=0}^{\infty} (-1)^n$ oscillates finitely.

Geometric Series:-

Geometric series are series of the form

$$a + ar + ar^2 + \dots + ar^{n-1} + \dots = \sum_{n=1}^{\infty} ar^{n-1}$$

in which a and r are fixed real numbers and $a \neq 0$.

The series can also be written as $\sum_{n=0}^{\infty} ar^n$.

If $r=1$, the n th partial sum of the geometric series is

$$S_n = a + a(1) + a(1)^2 + \dots + a(1)^{n-1} \\ = na$$

$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} na = \pm \infty$, depending on the sign of a . Therefore, the series diverges.

If $r = -1$, then $S_n = a - a + a - a + \dots + a(-1)^{n-1}$
 $= \begin{cases} a & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$

Since the n th partial sum S_n never approach a single limit and hence the series diverges.

If $|r| \neq 1$, we can determine the convergence or divergence of the series in the following way:

$$S_n = a + ar + ar^2 + \dots + ar^{n-1}$$

$$rS_n = ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n$$

$$S_n - rS_n = a - ar^n$$

$$S_n = \frac{a(1-r^n)}{1-r} \quad (|r| \neq 1)$$

If $|r| < 1$, then $r^n \rightarrow 0$ as $n \rightarrow \infty$

$$\text{Hence, } S_n \rightarrow \frac{a}{1-r}$$

In this case, the series converges and $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$

If $|r| > 1$, then $|r|^n \rightarrow \infty$ and the series diverges.

Finally, we conclude the following:

If $|r| < 1$, the geometric series $a + ar + ar^2 + \dots + ar^{n-1} + \dots$ converges to $\frac{a}{1-r}$

$$\text{i.e., } \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1$$

If $|r| \geq 1$, the series diverges.

② prove that the geometric series $1 + x + x^2 + \dots + x^{n-1} + \dots$

i) converges if $-1 < x < 1$ i.e., $|x| < 1$

ii) diverges if $|x| \geq 1$

Sol: please try yourself.

Example ① The geometric series with $a = \frac{1}{9}$ and $r = \frac{1}{3}$ is

$$\begin{aligned} \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \dots &= \sum_{n=1}^{\infty} \frac{1}{9} \left(\frac{1}{3}\right)^{n-1} = \frac{\frac{1}{9}}{1 - \frac{1}{3}} \\ &= \frac{1}{6} \end{aligned}$$

Example ② The geometric series with $a = 5$ and $r = -\frac{1}{4}$ is

$$\begin{aligned} 5 - \frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots &= \sum_{n=0}^{\infty} 5 \left(-\frac{1}{4}\right)^n \\ &= \frac{5}{1 + \frac{1}{4}} = 4 \end{aligned}$$

Ex 1 Determine if the geometric series converges or diverges. If a series converges, find its sum.

i) $1 - \frac{2}{e} + \left(\frac{2}{e}\right)^2 - \left(\frac{2}{e}\right)^3 + \left(\frac{2}{e}\right)^4 - \dots$

ii) $1 + \left(\frac{7}{6}\right)^2 + \left(\frac{7}{6}\right)^4 + \left(\frac{7}{6}\right)^6 + \left(\frac{7}{6}\right)^8 + \dots$

Example 3 Express the repeating decimal $5.\overline{23}$ (or) $5.232323\dots$ as the ratio of two integers.

Sol $5.232323\dots = 5 + \frac{23}{100} + \frac{23}{(100)^2} + \frac{23}{(100)^3} + \dots$

$$= 5 + \frac{23}{100} \left(1 + \frac{1}{100} + \left(\frac{1}{100}\right)^2 + \dots \right)$$

$$= 5 + \frac{23}{100} \cdot \frac{1}{1 - \frac{1}{100}} \quad \left(\because a = 1, r = \frac{1}{100} \right)$$

$$= 5 + \frac{23}{100} \times \frac{100}{99}$$

$$= 5 + \frac{23}{99} = \frac{518}{99}$$

Ex 2 Express the following repeating decimals as the ratio of two integers.

i) $0.\overline{37}$

ii) $1.\overline{732}$

Theorem If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$

(OR)

If a series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$

The converse of the above theorem is not always true.

i.e., $\lim_{n \rightarrow \infty} a_n = 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ may or may not be convergent.

Theorem:- The n th term test for divergence:

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \rightarrow \infty} a_n$ fails to exist or is different from zero.

Example:- i, $\sum_{n=1}^{\infty} n^r$ diverges because $\lim_{n \rightarrow \infty} n^r = \infty$

ii, $\sum_{n=1}^{\infty} \frac{n}{n+1}$ diverges because $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0$

iii, $\sum_{n=1}^{\infty} \frac{-n}{2n+7}$ diverges because $\lim_{n \rightarrow \infty} \frac{-n}{2n+7} = -\frac{1}{2} \neq 0$

iv, $\sum_{n=1}^{\infty} (-1)^{n+1}$ diverges because $\lim_{n \rightarrow \infty} (-1)^{n+1}$ does not exist.

Note ① $\sum_{n=1}^{\infty} a_n$ is convergent $\Rightarrow \lim_{n \rightarrow \infty} a_n = 0$

② $\lim_{n \rightarrow \infty} a_n = 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ may or may not be convergent

③ $\lim_{n \rightarrow \infty} a_n \neq 0$ (or) $\lim_{n \rightarrow \infty} a_n$ does not exist $\Rightarrow$
 $\sum_{n=1}^{\infty} a_n$ is divergent.

Example:- The series $\sum_{n=1}^{\infty} \frac{n+1}{n} = \frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \dots$

diverges because the partial sums eventually outgrow every preassigned number. Each term is ~~greater~~ greater than 1, so sum of n terms is greater than n .

* One reason that a series may fail to converge is that its terms don't become small.

Examples:- The n th term of the series may tend to zero as $n \rightarrow \infty$ (i.e., $\lim_{n \rightarrow \infty} a_n = 0$) but the series not convergent / divergent.

i) The series

$$1 + \underbrace{\frac{1}{2} + \frac{1}{2}}_{2 \text{ times}} + \underbrace{\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}}_{4 \text{ times}} + \dots + \underbrace{\frac{1}{2^n} + \frac{1}{2^n} + \dots + \frac{1}{2^n}}_{2^n \text{ times}}$$

+ ...

diverges even though $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{1}{2^n} + \frac{1}{2^n} + \dots + \frac{1}{2^n} \right) = 0$

ii) The series $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots$ diverges even though $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Theorem:- Suppose $\sum a_n$ and $\sum b_n$ are two convergent series and if $\sum a_n = A$ and $\sum b_n = B$, then

- i) Sum Rule: $\sum (a_n + b_n) = \sum a_n + \sum b_n = A + B$
- ii) Difference Rule: $\sum (a_n - b_n) = \sum a_n - \sum b_n = A - B$
- iii) Constant multiple Rule: $\sum k a_n = k \sum a_n = kA$ (for any number k)

Results: ① If $\sum a_n$ is divergent, then so is $\sum c a_n$ where $c \neq 0$
i.e., Every non zero constant multiple of a divergent series diverges.

② If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum (a_n + b_n)$ and $\sum (a_n - b_n)$ both diverges.

Note:- $\sum (a_n + b_n)$ can converge even if both $\sum a_n$ and $\sum b_n$ diverges.

For Example, $\sum a_n = 1 + 1 + 1 + \dots$ and
 $\sum b_n = (-1) + (-1) + (-1) + \dots$

both are diverges but $\sum (a_n + b_n) = 0 + 0 + 0 + \dots$

converges to 0

Example Find the sums of the following series.

i) $\frac{3}{5} + \frac{4}{5^2} + \frac{3}{5^3} + \frac{4}{5^4} + \dots$

ii) $\sum_{n=1}^{\infty} \frac{3^n - 1}{6^{n-1}}$

iii) $\sum_{n=0}^{\infty} \frac{4}{2^n}$

Sol i) $\left(\frac{3}{5} + \frac{3}{5^3} + \dots \right) + \left(\frac{4}{5^2} + \frac{4}{5^4} + \dots \right) = \sum a_n + \sum b_n$

Now $\sum a_n$ is a geometric series with common ratio $= \frac{1}{5^2}$ which is numerically less than 1, therefore $\sum a_n$ is convergent.

Similarly, $\sum b_n$ is also convergent.

$\therefore \sum (a_n + b_n)$ is also convergent.

$$\sum a_n + \sum b_n = \frac{3/5}{1 - \frac{1}{5^2}} + \frac{4/5}{1 - \frac{1}{5^2}}$$

$$= \frac{3}{5} \cdot \frac{25}{24} + \frac{4}{5} \cdot \frac{25}{24} = \frac{15}{24} + \frac{20}{24}$$

$$\begin{aligned} \text{ii) } \sum_{n=1}^{\infty} \frac{3^n - 1}{6^{n-1}} &= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} - \sum_{n=1}^{\infty} \left(\frac{1}{6}\right)^{n-1} \\ &= \frac{1}{1 - \frac{1}{2}} - \frac{1}{1 - \frac{1}{6}} = 2 - \frac{6}{5} = \frac{4}{5} \end{aligned}$$

$$\text{iii)} \quad \sum_{n=0}^{\infty} \frac{4}{2^n} = 4 \sum_{n=0}^{\infty} \frac{1}{2^n} \\ = 4 \left(\frac{1}{1-1/2} \right) = 8$$

Adding or Deleting terms: We can add a finite number of terms to a series or delete a finite number of terms without altering the series convergence or divergence, although in the case of convergence this will usually change the sum.

If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$

$$\text{and} \quad \sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots + a_{k-1} + \sum_{n=k}^{\infty} a_n.$$

Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$, then

$\sum_{n=1}^{\infty} a_n$ converges. Thus

$$\sum_{n=1}^{\infty} \frac{1}{3^n} = \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \sum_{n=5}^{\infty} \frac{1}{3^n}$$

$$\text{and} \quad \sum_{n=5}^{\infty} \frac{1}{3^n} = \left(\sum_{n=1}^{\infty} \frac{1}{3^n} \right) - \frac{1}{3} - \frac{1}{3^2} - \frac{1}{3^3} - \frac{1}{3^4}$$

The convergence or divergence of a series is not affected by its few terms. Only the tail of the series, ~~the~~ the part that remains when we sum beyond some finite

number of initial terms, influences whether it converges or diverges.

Conclusion:- The convergence or divergence of a series is not affected by addition or omission of a finite number of its terms.

Reindexing:- As long as we preserve the order of its terms, we can reindex any series without altering its convergence.

To raise the starting value of the index h units, replace the n in the formula for a_n by $n-h$:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1+h}^{\infty} a_{n-h} = a_1 + a_2 + a_3 + \dots$$

To lower the starting value of the index h units, replace the n in the formula for a_n by $n+h$:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1-h}^{\infty} a_{n+h} = a_1 + a_2 + a_3 + \dots$$

Note:- We usually give preference to indexings that lead to simple expressions.

For example, the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \dots$ can rewrite as $\sum_{n=0}^{\infty} \frac{1}{2^n}$, $\sum_{n=5}^{\infty} \frac{1}{2^{n-5}}$ or $\sum_{n=-3}^{\infty} \frac{1}{2^{n+3}}$.

Def: Series of Positive Terms

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If all the terms of the series $\sum a_n = a_1 + a_2 + a_3 + \dots$ are positive, i.e., if $a_n > 0 \forall n$, then the series $\sum a_n$ is called a series of positive terms.

Suppose that $\sum_{n=1}^{\infty} a_n$ is a series of positive terms (or non-negative terms) and let s_n be ~~the~~ its n th partial sum. Then $s_{n+1} = a_1 + a_2 + \dots + a_n + a_{n+1}$.

$$s_{n+1} = s_n + a_{n+1}$$

$$\Rightarrow s_{n+1} - s_n = a_{n+1} > 0 \forall n$$

$$\Rightarrow s_{n+1} > s_n \forall n$$

$\Rightarrow \{s_n\}$ is a monotonically increasing sequence.

Two cases arise. The sequence of partial sums $\{s_n\}$ may be bounded or unbounded above.

Case I: When $\{s_n\}$ is bounded above.

Since $\{s_n\}$ is monotonically increasing and bounded above, it is convergent $\Rightarrow \sum a_n$ is convergent.

Case II: When $\{s_n\}$ is not bounded above.

Since $\{s_n\}$ is monotonically increasing and not bounded above, it diverges to $+\infty \Rightarrow \sum a_n$ diverges to $+\infty$.

Theorem:- A series $\sum_{n=1}^{\infty} a_n$ of nonnegative terms/positive terms converges $\iff$ its partial sums are bounded from above.

Example:- Consider the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n} + \dots$$

the harmonic series is divergent because there is no upper bound for its partial sums.

$$1 + \frac{1}{2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{> \frac{2}{4} = \frac{1}{2}} + \underbrace{\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}}_{> \frac{4}{8} = \frac{1}{2}} + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \dots + \frac{1}{16}\right)}_{> \frac{8}{16} = \frac{1}{2}} + \dots$$

The sum of the first two terms is 1.5.

The sum of the next two terms $\frac{1}{3} + \frac{1}{4}$ is greater than $\frac{1}{2}$.

The sum of the next four terms $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > \frac{1}{2}$, and so on.

In general, the sum of 2^n terms ending with $\frac{1}{2^{n+1}}$ is greater than $\frac{1}{2}$.

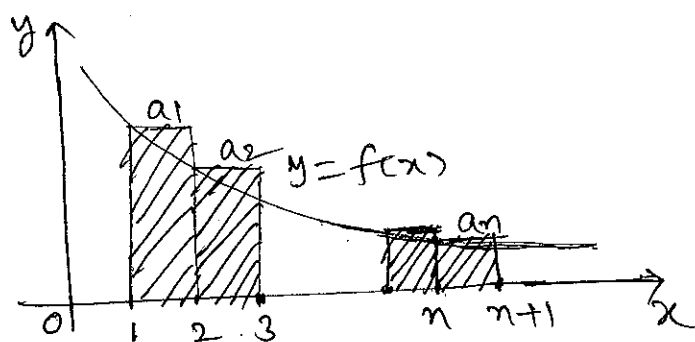
If $n = 2^k$, the partial sum s_n is greater than $\frac{k}{2}$, so the sequence of partial sums is not bounded from above. The harmonic series diverges.

Theorem:- The Integral Test

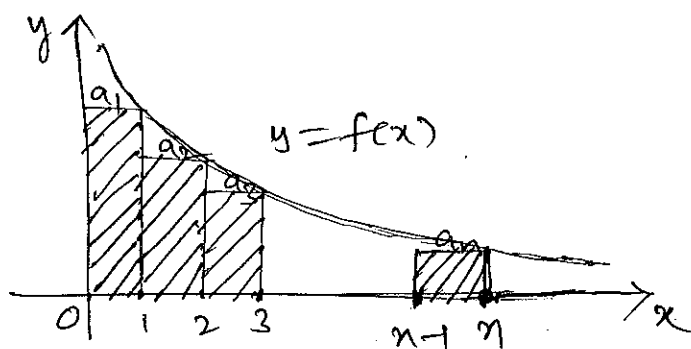
Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the improper integral $\int_N^{\infty} f(x) dx$ both converge or both diverge.

proof:- We establish the test for the case $N=1$. The proof for general N is similar.

Assume that $a_n = f(n) \forall n$



(a)



(b)

It is observed that the rectangles in Figure (a), which have areas $a_1, a_2, \dots, a_n$, collectively enclose more area than that under the curve $y = f(x)$ from $x=1$ to $x=n+1$. That is,

$$\int_1^{n+1} f(x) dx \leq a_1 + a_2 + a_3 + \dots + a_n.$$

In Figure (b) the rectangles have been faced to the left

Instead of to the right. If we momentarily disregard the first rectangle of area a_1 , we see that

$$a_2 + a_3 + \dots + a_n \leq \int_1^n f(x) dx$$

If we include a_1 , we have

$$a_1 + a_2 + a_3 + \dots + a_n \leq a_1 + \int_1^n f(x) dx$$

Hence,

$$\int_1^{n+1} f(x) dx \leq a_1 + a_2 + \dots + a_n \leq a_1 + \int_1^n f(x) dx$$

These inequalities hold for each n , and continue to hold as $n \rightarrow \infty$.

If $\int_1^{\infty} f(x) dx$ is finite, the right-hand side inequality shows that $\sum a_n$ is finite.

If $\int_1^{\infty} f(x) dx$ is infinite, the left-hand inequality shows that $\sum a_n$ is infinite. Hence the series and the integral are either both finite or both infinite.

Example: Does the following series converge?

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$$

Sol: We determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^2}$ by comparing

It with $\int_1^{\infty} \frac{1}{x^2} dx$.

To carry out the comparison, we think of the terms of the series as the values of the function $f(x) = \frac{1}{x^2}$ and interpret the values as the areas of rectangles under the curve

$$y = \frac{1}{x^2}$$

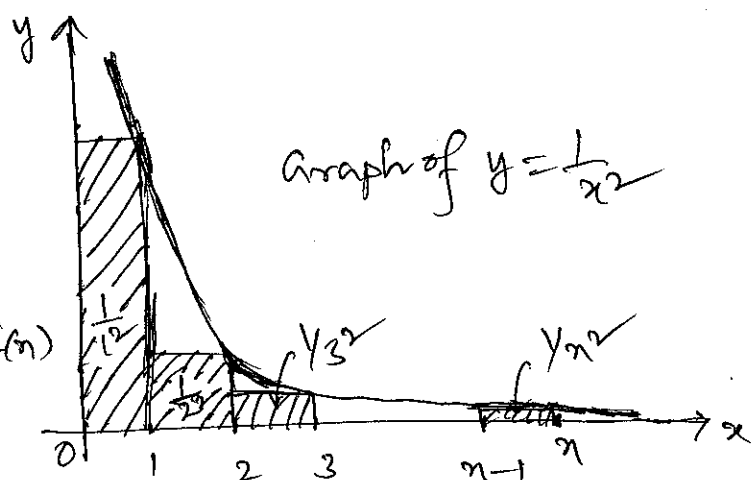
$$S_n = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}$$

$$= f(1) + f(2) + f(3) + \dots + f(n)$$

$$\bullet < f(1) + \int_1^n \frac{1}{x^2} dx$$

$$< 1 + \int_1^{\infty} \frac{1}{x^2} dx$$

$$< 1 + 1 = 2$$



i.e., the n th partial sum ~~is~~ S_n bounded above by 2

Therefore, the sequence of partial sums $\{S_n\}$ converges.

Hence the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Caution: The series and integral need not have the same value on the convergent case.

In the above example, $\sum_{n=1}^{\infty} \frac{1}{n^2} \neq \int_1^{\infty} f(x) dx$

($\because \int_1^{\infty} \frac{1}{x^2} dx = 1$, but $\sum_{n=1}^{\infty} \frac{1}{n^2} \neq 1$ and $\sum_{n=1}^{\infty} \frac{1}{n^2} < 2$)

Example 1:- Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$

Sol:- Let $a_n = \frac{1}{n^2+1}$

The function $f(x) = \frac{1}{x^2+1}$ is positive, continuous, and decreasing for all $x \geq 1$, and

$$\int_1^{\infty} f(x) dx = \int_1^{\infty} \frac{1}{x^2+1} dx$$

$$= \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^2+1} dx$$

$$= \lim_{b \rightarrow \infty} [\tan^{-1} x]_1^b$$

$$= \lim_{b \rightarrow \infty} (\tan^{-1} b - \tan^{-1} 1)$$

$$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \text{ (finite)}$$

The Integral Test tells us that the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges, but it does not say that $\sum_{n=1}^{\infty} \frac{1}{n^2+1} = \frac{\pi}{4}$.

Example 2:- Determine the convergence or divergence of the

series (a) $\sum_{n=1}^{\infty} n e^{-n^2}$

(b) $\sum_{n=1}^{\infty} \frac{1}{2 \ln n}$

Sol: (a) We apply the Integral Test

$$\int_1^{\infty} \frac{x}{e^{x^2}} dx = \int_1^{\infty} \frac{dt/2}{e^t}$$

$$\begin{aligned} x^2 &= t \\ 2x dx &= dt \\ x dx &= dt/2 \end{aligned}$$

$$= \frac{1}{2} \lim_{b \rightarrow \infty} \int_1^b \frac{1}{e^t} dt$$

$$= \frac{1}{2} \cdot \cancel{\infty} \lim_{b \rightarrow \infty} \left[-e^{-t} \right]_1^b$$

$$= \frac{1}{2} \lim_{b \rightarrow \infty} \left[-e^{-b} + e^{-1} \right]$$

$$= \frac{1}{2e} \text{ (finite)}$$

Since the integral converges, the series also converges.

(b) Apply the Integral Test

$$\int_1^{\infty} \frac{1}{2 \ln x} dx = \int_0^{\infty} \frac{e^t dt}{2^t}$$

$$\begin{aligned} \text{let } \ln x &= t \\ \frac{1}{x} dx &= dt \\ dx &= x dt \\ &= e^t dt \end{aligned}$$

$$= \int_0^{\infty} \left(\frac{e}{2} \right)^t dt$$

$$= \lim_{b \rightarrow \infty} \int_0^b \left(\frac{e}{2} \right)^t dt$$

$$= \lim_{b \rightarrow \infty} \frac{1}{\ln(\frac{e}{2})} \left[\left(\frac{e}{2} \right)^b - \left(\frac{e}{2} \right)^0 \right]$$

$$= \infty$$

The Improper Integral diverges, so the series diverges.

* p-series Test:- The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} + \dots$ (p a real constant)

converges if $p > 1$, and diverges if $p \leq 1$.

Example: show that the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} + \dots$$

converges if $p > 1$ and diverges if $p \leq 1$.

Sol: Here $a_n = \frac{1}{n^p} = f(n)$

$$\therefore f(x) = \frac{1}{x^p}$$

If $p > 1$, then $f(x) = \frac{1}{x^p}$ is a positive decreasing function of x .

$\therefore$ The Integral Test is applicable.

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^p} dx &= \int_1^{\infty} x^{-p} dx = \lim_{b \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_1^b \\ &= \left(\frac{1}{1-p} \lim_{b \rightarrow \infty} \left(\frac{1}{b^{p-1}} \right) - 1 \right) \left(\because \begin{array}{l} p-1 > 0 \\ b \rightarrow \infty \Rightarrow \frac{1}{b^{p-1}} \rightarrow 0 \end{array} \right) \\ &= \frac{1}{1-p} (0 - 1) = \frac{1}{p-1} \text{ (finite)} \end{aligned}$$

$\therefore$ The series converges by Integral Test. Note that the sum of the p-series is not $\frac{1}{p-1}$. The series converges,

but we don't know the value it converges to.

(20)

If $p=1$, we have the harmonic series

$$\sum \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots$$

which is divergent.

If $0 < p < 1$, then $1-p > 0$ and

$$\int_1^{\infty} \frac{1}{x^p} dx = \frac{1}{1-p} \lim_{b \rightarrow \infty} (b^{1-p} - 1) \\ = \infty$$

therefore, the series diverges by Integral Test

If $p \leq 0$, the series diverges by the n th term test.

Hence $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Ex 1 Using Integral Test, show that the series

$$\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^p}, \quad p > 0 \text{ converges if } p > 1, \text{ and diverges}$$

if $0 < p \leq 1$.

Sol: - please try yourself.

Ex 2 Use Integral Test to determine convergence or divergence of the following series:

$$(a) \sum_{n=1}^{\infty} \frac{1}{n+4} \quad (b) \sum_{n=1}^{\infty} \frac{n}{n^2+1} \quad (c) \sum_{n=1}^{\infty} \frac{8 \tanh n}{1+n^2}$$

* We have two questions on a series $\sum a_n$:

i) Does the series converge?

ii) If it converges, what is its sum?

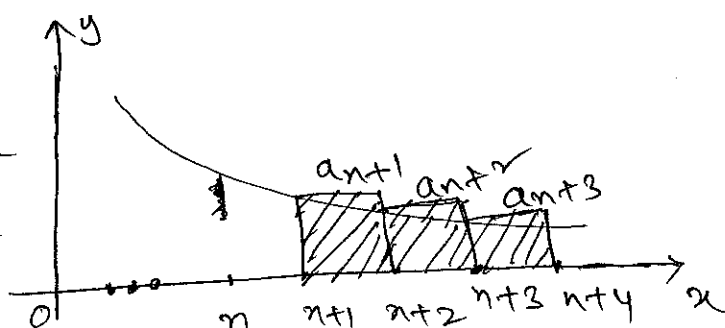
Error estimation: For some convergent series, such as the geometric series or the telescoping series, we can actually find the total sum of the series. That is, we can find the limiting value S of the sequence of partial sums. For most convergent series, however, we cannot easily find the total sum. Nevertheless, we can estimate the sum by adding the first n terms to get s_n , but we need to know how far off s_n is from the total sum S .

An approximation to a function or to a number is more useful when it is accompanied by a bound on the size of the worst possible error that could occur. With such an error bound we can try to make an estimate or approximation that is close enough for the problem at hand. Without a bound on the error size, we are just guessing and hoping that we are close to the actual answer. We now show a way to bound the error size using integrals.

Suppose that a series $\sum a_n$ with positive terms is shown to be convergent by the Integral Test, and we want to estimate the size of the remainder R_n measuring between the total sum S of the series and its n th partial sum S_n . That is, we wish to estimate

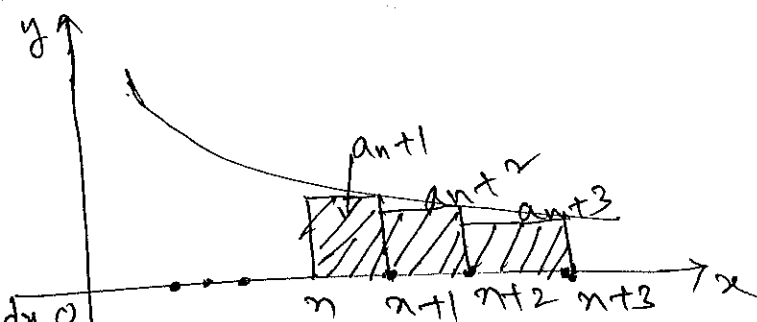
$$R_n = S - S_n = a_{n+1} + a_{n+2} + a_{n+3} + \dots$$

To get a lower bound for the remainder, we compare the sum of the areas of the rectangles with the area under the curve $y = f(x)$ for $x \geq n$.



We see that

$$R_n = a_{n+1} + a_{n+2} + \dots \geq \int_{n+1}^{\infty} f(x) dx$$



Similarly, we find an upper bound with

$$R_n = a_{n+1} + a_{n+2} + a_{n+3} + \dots \leq \int_n^{\infty} f(x) dx$$

Bounds for the remainder in the Integral Test

Suppose $\{a_n\}$ is a sequence of positive terms $a_n = f(n)$, where f is continuous positive decreasing function

of x for all $x \geq n$, and that $\sum a_n$ converges to s .

Then the remainder $R_n = s - s_n$ satisfies the inequalities

$$\boxed{\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx} \rightarrow \textcircled{1}$$

$$\text{i.e., } s_n + \int_{n+1}^{\infty} f(x) dx \leq s \leq s_n + \int_n^{\infty} f(x) dx \rightarrow \textcircled{2}$$

The inequalities in $\textcircled{2}$ are useful for estimating the error in approximating the sum of a series known to converge by the Integral Test. The error can be no larger than the length of the interval containing s , with endpoints given by $\textcircled{2}$.

Example: Estimate the sum of the series $\sum \frac{1}{n^2}$ using the inequality in $\textcircled{2}$ and $n=10$.

Sol.: $\int_n^{\infty} f(x) dx = \int_n^{\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{x} \right]_n^b = \frac{1}{n}$

$$\int_{10}^{\infty} f(x) = \frac{1}{10} \quad \text{and also} \quad \int_{11}^{\infty} f(x) dx = \frac{1}{11}$$

$$\text{from } \textcircled{2}, \quad s_{10} + \frac{1}{11} \leq s \leq s_{10} + \frac{1}{10} \rightarrow \textcircled{*}$$

Taking $s_{10} = 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{100} \approx 1.54977$

$$(*) \Rightarrow 1.64068 \leq S \leq 1.64977$$

If we approximate the sum S by the midpoint of this interval, we find that

$$S = \sum_{n=1}^{\infty} \frac{1}{n^2} \approx 1.6452$$

The error in this approximation is less than half the length of the interval, so the error is less than 0.005.

Note:- Using trigonometric Fourier series, it can be shown that $S = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.64493$.

Cauchy's Condensation Test:-

Let $\{a_n\}$ be a nonincreasing sequence ($a_n \geq a_{n+1} \forall n$) of positive terms that converges to 0. Then $\sum a_n$ and $\sum 2^n a_{2^n}$ both converge or both diverge.

For example, $\sum \frac{1}{n}$ diverges because $\sum 2^n \frac{1}{2^n} = \sum 1$ diverges.

For example, $\sum \frac{1}{n^2}$ converges because $\sum 2^n \frac{1}{(2^n)^2} = \sum \frac{1}{2^n}$ converges.

Example Use Cauchy condensation test, discuss the convergence of the following series:

i) $\sum_{n=2}^{\infty} \frac{1}{\ln n}$

ii) $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

iii) $\sum_{n=2}^{\infty} \frac{1}{n \sqrt{\ln n}}$ or $\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^{1/2}}$

iv) $\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^{3/2}}$

v) $\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^p}, p > 0$

vi) $\sum_{n=1}^{\infty} \frac{1}{n^p}$

vii) $\sum_{n=1}^{\infty} \frac{\ln n}{n}$

Solr i) let $a_n = \frac{1}{\ln n}$

Since $\{\ln n\}$ is increasing sequence, $\{a_n\}$ is decreasing sequence i.e., $a_n > a_{n+1} \forall n \geq 2$

$$\begin{aligned} \text{Now } \sum_{n=2}^{\infty} 2^n a_{2^n} &= \sum_{n=2}^{\infty} 2^n \frac{1}{\ln 2^n} = \sum_{n=2}^{\infty} \frac{2^n}{n \ln 2} \\ &= \frac{1}{\ln 2} \sum_{n=2}^{\infty} \frac{2^n}{n} \end{aligned}$$

$$\lim_{n \rightarrow \infty} \frac{2^n}{n} = \lim_{n \rightarrow \infty} \frac{2^n \cdot \ln 2}{1} = \infty \text{ and therefore,}$$

$$\sum_{n=2}^{\infty} \frac{2^n}{n} \text{ diverges}$$

$$\Rightarrow \frac{1}{\ln 2} \sum_{n=2}^{\infty} \frac{2^n}{n} \text{ diverges}$$

$$\therefore \sum_{n=2}^{\infty} \frac{2^n}{2^n} a_{2^n} \text{ diverges}$$

$\therefore$ By Cauchy's condensation test, the series $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} \frac{2^n}{2^n} a_{2^n}$ converge or diverge together.

$$\text{Hence } \sum_{n=2}^{\infty} \frac{1}{n \ln n} \text{ diverges.}$$

(ii) Here $a_n = \frac{1}{n \ln n}$

Since $\{n \ln n\}$ is an increasing sequence, $\{a_n\}$ is a decreasing sequence and converges to 0.

$$\text{Now } \sum_{n=2}^{\infty} 2^n a_{2^n} = \sum_{n=2}^{\infty} 2^n \cdot \frac{1}{2^n \cdot \ln 2^n} = \sum_{n=2}^{\infty} \frac{1}{n \ln 2}$$

$$= \frac{1}{\ln 2} \sum_{n=2}^{\infty} \frac{1}{n}$$

Since $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges by p-series test and hence

$$\frac{1}{\ln 2} \sum_{n=2}^{\infty} \frac{1}{n} \text{ ~~converges~~ diverges.}$$

$$\therefore \sum_{n=2}^{\infty} \frac{2^n}{2^n} a_{2^n} \text{ diverges}$$

By Cauchy's condensation test $\sum_{n=2}^{\infty} a_n = \sum_{n=2}^{\infty} \frac{1}{n \ln n}$
~~con~~ diverges.

iii) please try yourself.

iv) please try yourself.

v) Here $a_n = \frac{1}{n(\ln n)^p}$, $p > 0$

Since $\{n(\ln n)^p\}$ is an increasing sequence, $\{a_n\}$ is a decreasing sequence and $a_n \rightarrow 0$ as $n \rightarrow \infty$.

$$\begin{aligned} \text{Now } \sum_{n=2}^{\infty} \frac{2^n}{2^n} a_{2^n} &= \sum_{n=2}^{\infty} \frac{2^n}{2^n} \cdot \frac{1}{2^n (\ln 2^n)^p} \\ &= \sum_{n=2}^{\infty} \frac{1}{(\ln 2)^p} \cdot \frac{1}{n^p} \\ &= \frac{1}{(\ln 2)^p} \sum_{n=2}^{\infty} \frac{1}{n^p} \end{aligned}$$

Since $\sum_{n=2}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

$$\therefore \frac{1}{(\ln 2)^p} \sum_{n=2}^{\infty} \frac{1}{n^p} \text{ converges if } p > 1 \text{ and diverges if } p \leq 1$$

$$\text{i.e., } \sum_{n=2}^{\infty} \frac{2^n}{2^n} a_{2^n} \text{ converges if } p > 1 \text{ and diverges if } p \leq 1.$$

(34)

$\therefore$ By Cauchy's condensation test, $\sum_{n=2}^{\infty} a_n$ converges
if $p > 1$ and diverges if $p \leq 1$

vi, please try yourself.

vii, please try yourself.

Comparison Tests:- We have seen how to determine the convergence of geometric series, p-series, and a few others. We can test the convergence of many more series by comparing their terms to those of a series whose convergence is already known.

Theorem: Direct Comparison Test

Let $\sum a_n$ and $\sum b_n$ be two series with $0 \leq a_n \leq b_n$ for all n . Then

i) If $\sum b_n$ converges, then $\sum a_n$ also converges.

ii) If $\sum a_n$ diverges, then $\sum b_n$ also diverges.

Example Use Direct comparison test to determine if each series converges or diverges.

i)
$$\sum_{n=1}^{\infty} \frac{5}{5n-1}$$

$$(i) \sum_{n=1}^{\infty} \frac{1}{n^2+30}$$

$$(iii) \sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}$$

$$(iv) \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$(v) \sum_{n=1}^{\infty} \frac{1}{2^n + \sqrt{n}}$$

Solⁿ: Here $a_n = \frac{5}{5n-1} = \frac{1}{n-\frac{1}{5}} > \frac{1}{n}$

Since $\sum \frac{1}{n}$ diverges and hence by comparison test,

$$\sum_{n=1}^{\infty} \frac{1}{n-\frac{1}{5}} \text{ or } \sum \frac{5}{5n-1} \text{ diverges.}$$

(ii) Here $a_n = \frac{1}{n^2+30}$

$$\therefore n^2+30 > n^2$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent by p-series test. $\Rightarrow \frac{1}{n^2+30} < \frac{1}{n^2}$

$$\text{Since } \frac{1}{n^2+30} < \frac{1}{n^2} \quad \forall n$$

Therefore, $\sum_{n=1}^{\infty} \frac{1}{n^2+30}$ is convergent.

(iii) please try your self

(iv) please try yourself.

(v) $a_n = \frac{1}{2^n + \sqrt{n}}$

Since $\frac{1}{2^n + \sqrt{n}} < \frac{1}{2^n}$

The series $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is converges and therefore by comparison test, the series $\sum_{n=1}^{\infty} \frac{1}{2^n + \sqrt{n}}$ converges.

The limit comparison test: The limit comparison test is particularly useful for series in which a_n is a rational function of n .

Theorem: Limit Comparison Test.

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N a positive integer).

i) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ and $c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.

ii) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

iii) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Example: Which of the following series converge, and which diverge?

$$i) \frac{3}{4} + \frac{5}{9} + \frac{7}{16} + \frac{9}{25} + \dots = \sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{2n+1}{n^2+2n+1}$$

$$ii) \sum_{n=1}^{\infty} \sqrt{\frac{n+1}{n^2+2}}$$

$$iii) \sum_{n=1}^{\infty} \frac{n-2}{n^3-n^2+3}$$

$$iv) \sum_{n=2}^{\infty} \frac{n(n+1)}{(n^2+1)(n-1)}$$

$$v) \sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n^2}\right)$$

$$vi) \frac{1}{3 \cdot 7} + \frac{1}{4 \cdot 9} + \frac{1}{5 \cdot 11} + \frac{1}{6 \cdot 13} + \dots$$

$$vii) \sum_{n=1}^{\infty} (\sqrt{n^3+1} - \sqrt{n^3})$$

$$viii) \sum (\sqrt{n^2+1} - n)$$

$$ix) \sum (\sqrt{n^2+1} - \sqrt{n^2-1})$$

$$x) \sum_{n=1}^{\infty} \frac{1}{1+2+3+\dots+n}$$

$$xi) \sum_{n=1}^{\infty} \frac{1}{1^2+2^2+3^2+\dots+n^2}$$

Sol: i) Here $a_n = \frac{2n+1}{n^2+2n+1}$

For large n , we expect a_n to behave like $\frac{2n}{n^2} = \frac{2}{n}$

since the leading terms dominate for large n , so we

let $b_n = \frac{1}{n}$. Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{2n^2 + n}{n^2 + 2n + 1} = 2,$$

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{2n+1}{n^2+2n+1} \text{ diverges, by } \begin{matrix} \text{Limit} \\ \uparrow \\ \text{Comparison} \end{matrix}$$

Test.

Sol: ii) Here $a_n = \sqrt{\frac{n+1}{n^2+2}}$

$$= \sqrt{\frac{n(1+\frac{1}{n})}{n^2(1+\frac{2}{n^2})}} = \frac{1}{n^{1/2}} \frac{(1+\frac{1}{n})}{(1+\frac{2}{n^2})}$$

Let us take $b_n = \frac{1}{n^{1/2}}$

Now $\frac{a_n}{b_n} = \frac{\frac{1}{n^{1/2}} \frac{(1+\frac{1}{n})}{(1+\frac{2}{n^2})}}{\frac{1}{n^{1/2}}}$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1+\frac{1}{n}}{1+\frac{2}{n^2}} = \frac{1+0}{1+0} = 1$$

which is finite and $\neq 0$

Therefore, by Limit comparison test, $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge or both diverge.

Since $b_n = \frac{1}{n^{1/2}}$ and $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ diverges by p-series test and hence $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{\sqrt{n+1}}{\sqrt{n^2+2}}$ diverges.

Sol iii), Here $a_n = \frac{n-2}{n^3-n^2+3}$

$$= \frac{n(1-2/n)}{n^3(1-\frac{1}{n}+\frac{3}{n^3})}$$

$$= \frac{1}{n^2} \frac{(1-2/n)}{(1-\frac{1}{n}+\frac{3}{n^3})}$$

let us take $b_n = \frac{1}{n^2}$, Now $\frac{a_n}{b_n} = \frac{1-2/n}{(1-\frac{1}{n}+\frac{3}{n^3})}$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{(1-2/n)}{(1-\frac{1}{n}+\frac{3}{n^3})} = \frac{1-0}{1-0+0} = 1 \text{ which}$$

is finite and non-zero

and $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ convergent by p-series test.

Therefore, the series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{n-2}{n^3-n^2+3}$ convergent.

Sol (iv) - (xi)

please try yourself.

Example: Test the convergence of the series:

i) $\sum \sin \frac{1}{n}$

ii) $\sum \frac{1}{n} \sin \frac{1}{n}$

iii) $\sum \frac{1}{\sqrt{n}} \cdot \sin \frac{1}{n}$

iv) $\sum \frac{1}{\sqrt{n}} \tan \frac{1}{n}$

v) $\sum \tan^{-1} \frac{1}{n}$

vi) $\sum \sin\left(\frac{1}{n^2}\right)$

Sol: i) Here $a_n = \sin \frac{1}{n}$

$$= \frac{1}{n} - \frac{1}{3!} \left(\frac{1}{n}\right)^3 + \frac{1}{5!} \left(\frac{1}{n}\right)^5 - \dots$$

$$= \frac{1}{n} \left[1 - \frac{1}{3!} \cdot \frac{1}{n^2} + \frac{1}{5!} \cdot \frac{1}{n^4} - \dots \right]$$

Let us take $b_n = \frac{1}{n}$

Now $\frac{a_n}{b_n} = 1 - \frac{1}{3!} \cdot \frac{1}{n^2} + \frac{1}{5!} \cdot \frac{1}{n^4} - \dots$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{3!} \cdot \frac{1}{n^2} + \frac{1}{5!} \cdot \frac{1}{n^4} - \dots \right)$$

$$= 1 - 0 + 0 - 0 - \dots = 1 \text{ which is finite and non-zero.}$$

By Limit Comparison Test, $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge or both diverge.

Since $\sum b_n = \sum \frac{1}{n}$ divergent and hence $\sum a_n = \sum \sin \frac{1}{n}$ divergent.

Sol. (ii) - $\sqrt{15}$

please try yourself.

Example Does the series $\sum_{n=1}^{\infty} \frac{1+n \ln n}{n^r+5}$ converge?

Sol. Here $a_n = \frac{1+n \ln n}{n^r+5}$

For large n , we expect a_n to behave like $\frac{n \ln n}{n^r} = \frac{\ln n}{n}$,

which is greater than $\frac{1}{n}$ for $n \geq 3$.

so we let $b_n = \frac{1}{n}$

$$\text{Now } \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n + n^r \ln n}{n^r + 5}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{\ln n + \frac{1}{n}}{1 + \frac{5}{n^r}} \right) = \frac{\infty + 0}{1 + 0} = \infty$$

$$\text{and } \sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges.}$$

$$\therefore \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1+n \ln n}{n^r+5} \text{ divergent by Limit Comparison Test.}$$

Note: If $\sum a_n$ is a convergent series of nonnegative terms, then $\sum a_n^r$ converges. Converse of this result need not be true.